

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 2
Subject: 340152 Electrical Measuring Instruments
Date: 12 March 2014
Year: 2013
Section: 5-6
Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. This is be a close-book test.
4. Calculation devices are allowed.
5. Electronic communication devices are NOT allowed in the examination room.
6. The examination has 8 pages (including this page), and a total score of 70 points.
7. Write all your answers on this examination sheet.

Q1) Explain their operation of two Voltmeter and Ammeter methods of resistance measurement and also sketch their circuit diagrams. (5 points)

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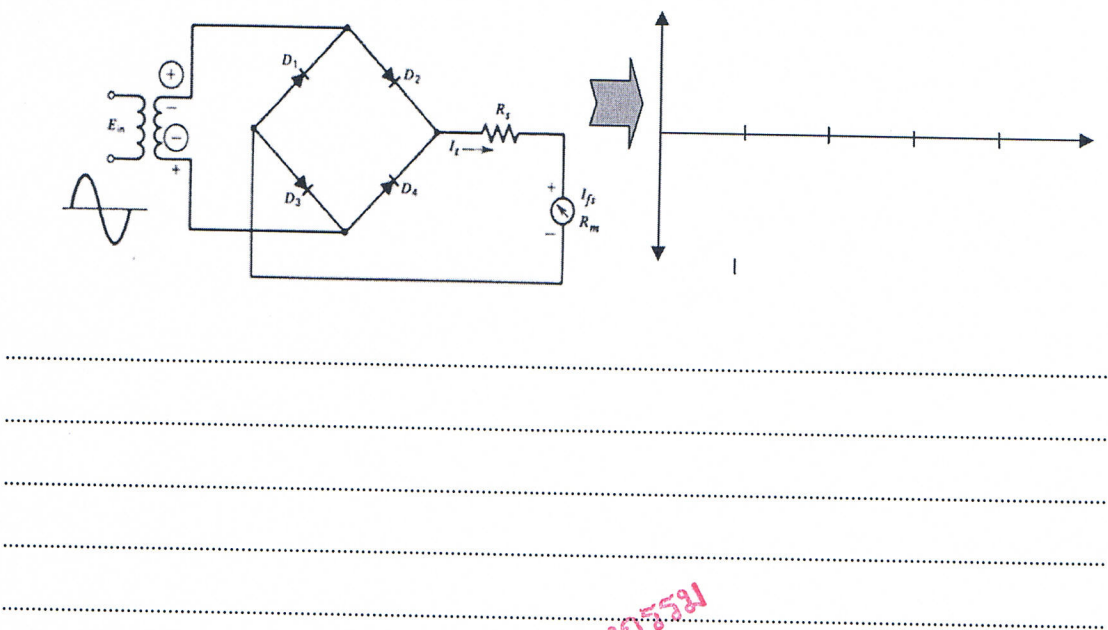
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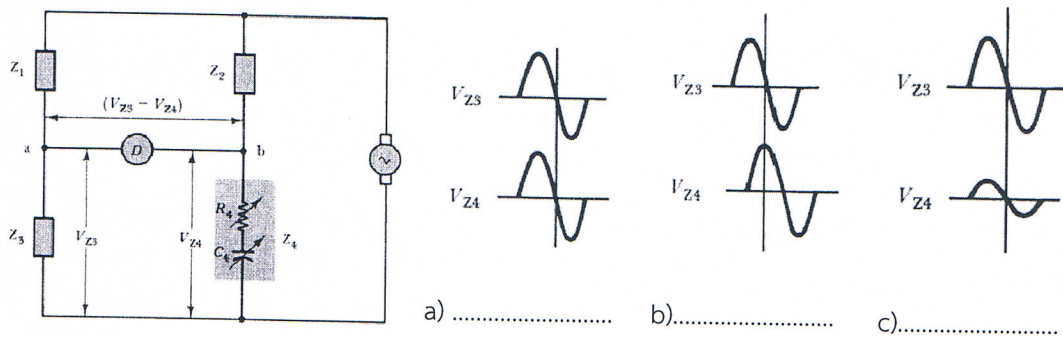
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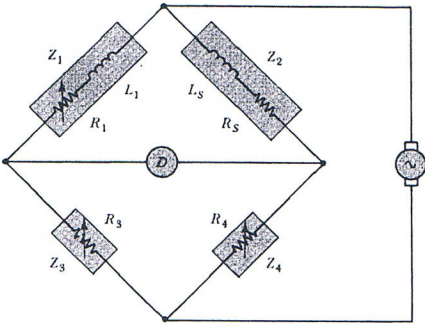
Q2) From the figure, **compute** the value of the multiplier resistor (R_s) for a 30-Vrms ac range on the voltmeter, where a PMMC instrument with FSD (I_{fs})= 500 μ A and meter coil resistance (R_m) is 600 Ω and **sketch** the voltage across the meter movement. (5 points)



Q3) From the figures, identify each of the case of AC bridge **balanced** or **unbalanced**. (3 points)



Q4) From a simple inductance bridge, derive the equations for the resistive (R_s) and inductive components (L_s). (5 points)



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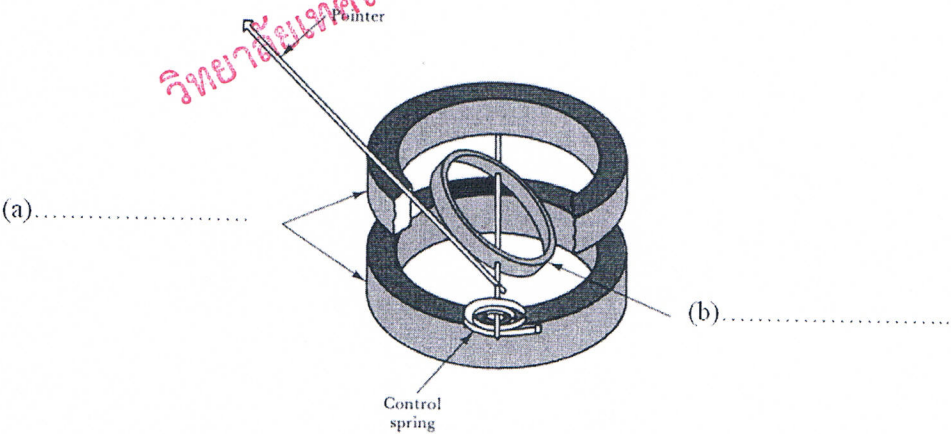
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Q5) Identify each part of the electrodynamic instrument and give the major advantage compare with PMMC. (4 points)



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Q6) Draw the basic electrodynamic meter as a Wattmeter. (3 points)

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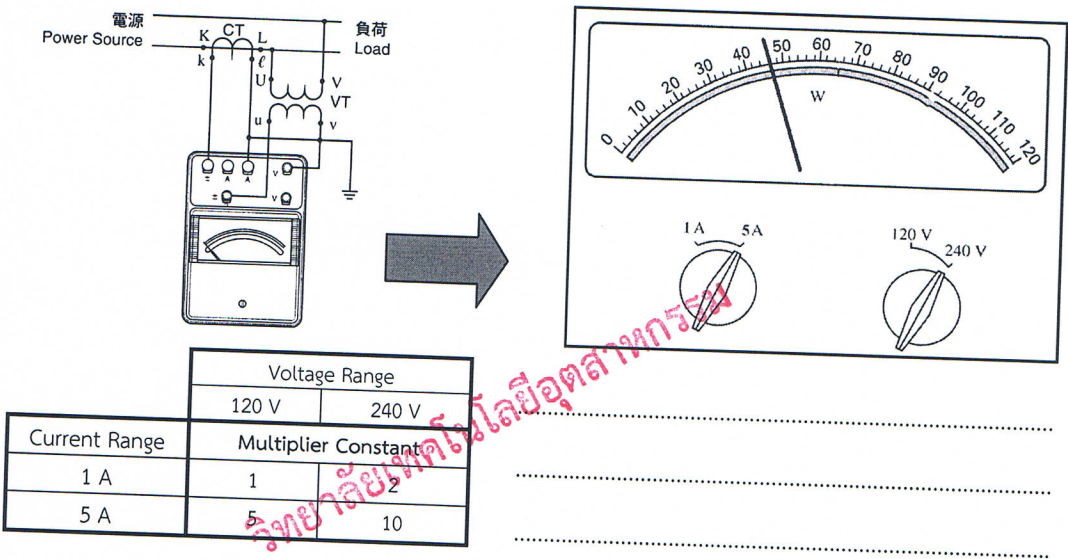
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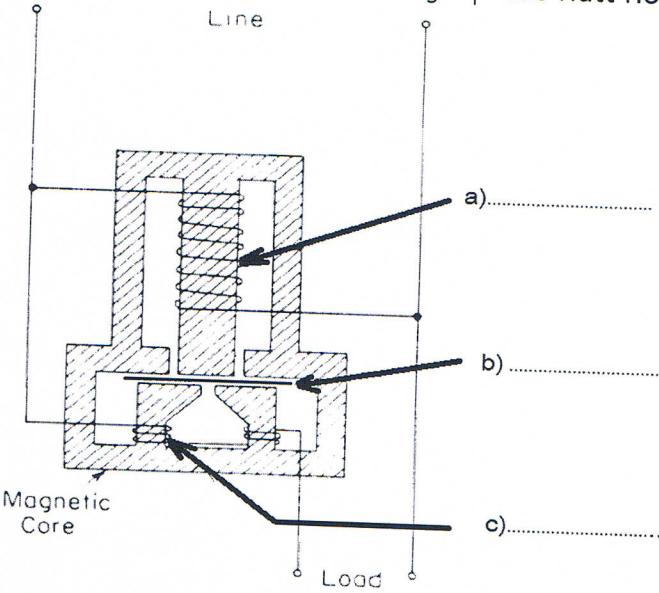
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Q7) For the wattmeter, CT, and VT in Figure, determine the measured power (watt) where: VT ratio = 6600/240 V, CT ratio = 100/5 A, respectively. (4 points)



Q8) Identify each part of the single phase watt-hour-meter. (3 points)



Q9) Describe the basic concept of Power factor meters.

(4 points)



Q10) List main functions of CT and PT.

(3 points)

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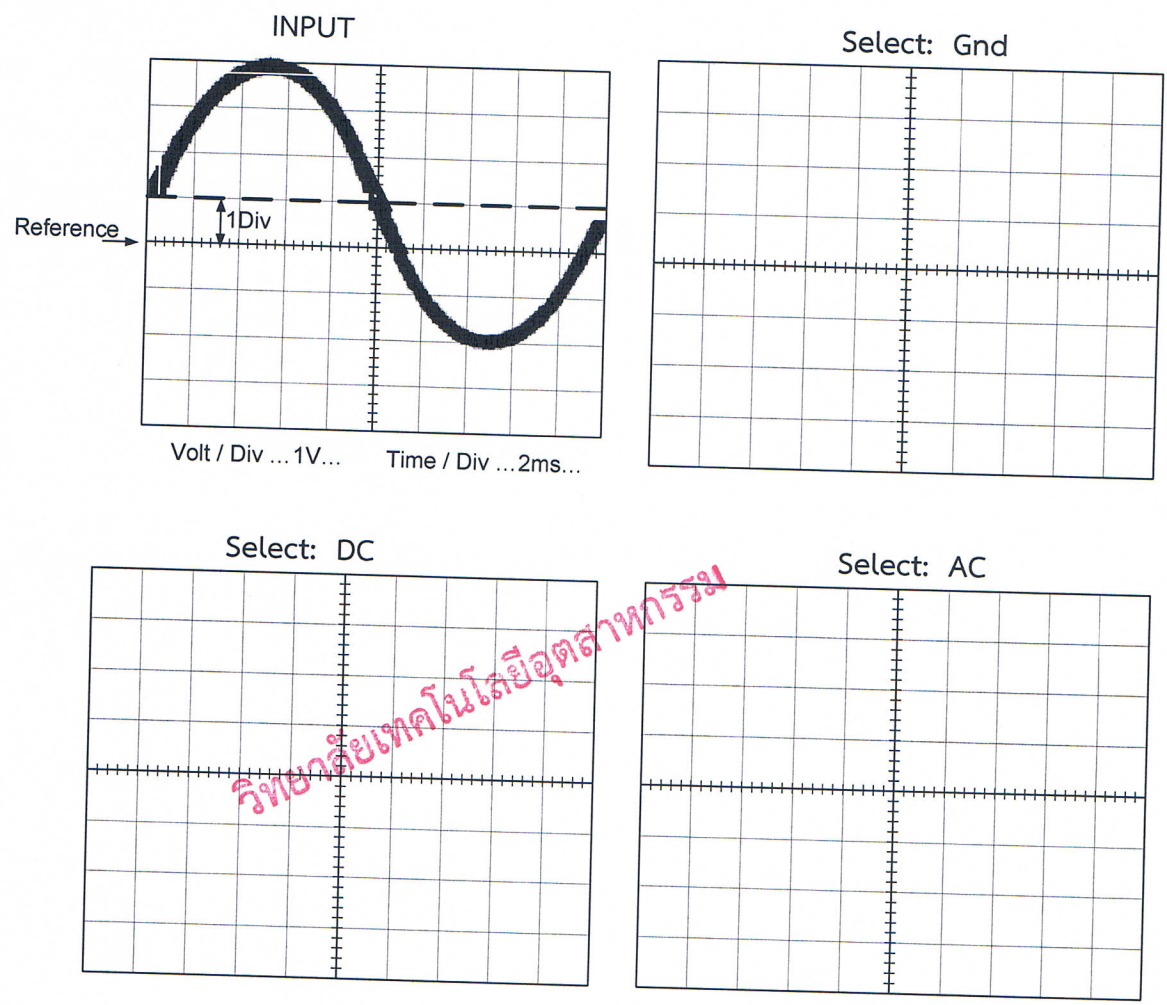
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Q11) Draw and explain the simplified block diagram of both *analog* and *digital* oscilloscopes.

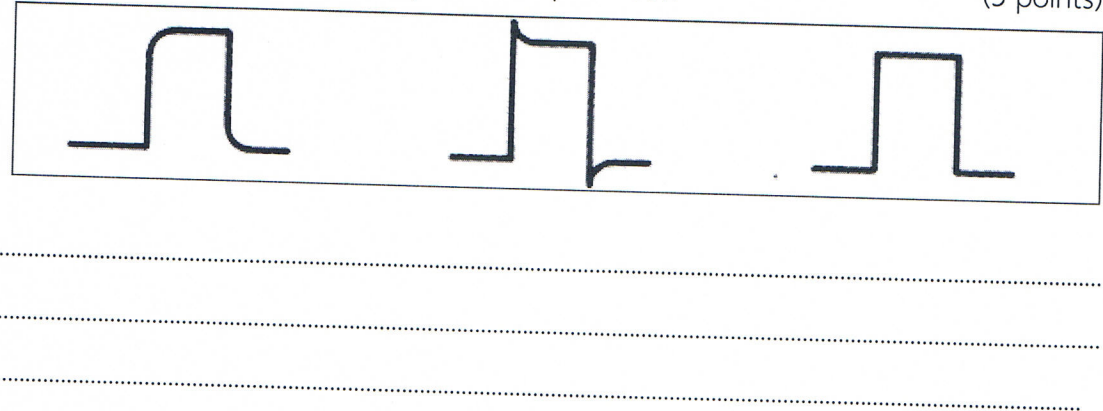
(7 points)

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Q14) From the input signal of the oscilloscope, draw the scope display when the setting of input coupling selector to Gnd, DC, and AC. (3 points)



Q15) Describe three figures of probe compensation (3 points)



Q16) Draw the **basic function generator** block diagram. (3 points)

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Q17) Show how to calibrate an **analog Wattmeter**. (4 points)

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Q18) Draw the **basic pulse generator** block diagram. (3 points)

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